3. Heart + · Write [DEM] The equivalence " $x^2 = 4 \Leftrightarrow x = 2$" is it correct? Correct.

Rep. No: $x = -2$ also fits. We have the involvement $x = 2 \Rightarrow x^2 = 4$ and equivalence $x^2 = 4 \Leftrightarrow (x = 2$ or $x = -2)$.

4. Back · Flash [CMP] Compare 2^{10} and 10^3.

Rep. $2^{10} = 1\,024 > 1\,000 = 10^3$.

5. Red thread · Flash [REP] Among which consecutive integers $\frac{7}{4}$ on a graduated right? decimal values.

Rep. Between 1 and 2; 1.75.

Form 61 Mon. 26 Apr. 2027 Common Oral: Comparison of Methods

1. Warming · Flash [GEO] Distance between $A(-2;1)$ and $B(1;5)$?

Rep. $\sqrt{3^2 + 4^2} = 5$.

2. Heart · Method [GEO] Identify three methods to prove that A, B, C are aligned.

Rep. (i) $\det\left(\overrightarrow{AB}, \overrightarrow{AC}\right) = 0$; (ii) equation of (AB) verified by C ; (iii) $AB + BC = AC$: equal distances, equivalent to $B \in [AC]$.

3. Heart + · Reflection [GEO] To prove that two lines given by $y = m_1 x + p_1$ and $y = m_2 x + p_2$ are parallel, what is the fastest method?

Rep. Compare the slopes: $m_1 = m_2$; parallels strictly if more $p_1 \neq p_2$.

4. Return · Factorize [DEV] Factor $6x + 9$.

Rep. $3(2x + 3)$.

5. Red thread · Method [IVD] Decompose 360 into prime factors. How many divisors does it have?

Rep. $360 = 2^3 \times 3^2 \times 5$; $(3+1)(2+1)(1+1) = 24$ divisors.

Problems of synthesis, reasoning and fluency

Form 62 Mar. 27 Apr. 2027 Synthesis problems crossing several fields of the programme

1. Warming · Flash [EQ] Solve $5x - 3 = 2x + 9$.

Rep. $3x = 12$, $x = 4$.

2. Heart · Problem [PRO] A price rises by 20% and then a fixed discount of 1 000 Ar and is then worth 11,000 Ar. What was the original price?

Rep. $1{,}2p - 1\,000 = 11\,000$, $p = \frac{12\,000}{1{,}2} = 10\,000$ Ar.

3. Heart + · Method [REG] $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the intersection point of the curves then solve $f(x) < g(x)$.

Rep. $2x + 1 = -x + 7$: $x = 2$, point $(2;5)$; $f(x) < g(x) \Leftrightarrow 3x < 6 \Leftrightarrow x < 2$.

4. Back · Flash [QP] A recipe for 4 people requires 250 g of flour. How much for 6 People?

Rep. $250 \div 4 \times 6 = 375$ g.

5. Red thread · Flash [GEO] A figure of 4 cm ^{2} is enlarged with a ratio 3. enlargement?

Rep. Area $\times 3^2$: 36 cm^2.

Trap: multiply the area by 3 for 3^2.